



Spot an Error

And WIN Award!

You can win one thousand rupees plus a certificate from the editor of Electronics For You magazine if you spot and report any factual error in the latest issue of the magazine. Your report will be published in a coming issue of the magazine, giving credit to you.

Though every care is taken in preparation of the content for the magazine, an error can slip in at times. So we invite our learned readers to point out any error that they spot, for the benefit of the other readers.

Rules:

- The error has to be reported before end of the month that is mentioned on cover of the magazine
- The report has to be sent in writing or by email addressed to the Editor, EFY Magazine (editsec@efyindia.com)
- The error should be of factual nature, in the information published in the magazine, and not a grammatical or syntax error
- The correct information should also be provided along with the error spotted by you
- In case more than one person reports the same error, the one whose report reaches us the earliest will get the award money
- In case two or more persons jointly report an error, the award money will be divided amongst them
- EFY employees, their family members and friends, and EFY associates and their family members and friends are not eligible
- Decision of the editor will be final in this matter

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to decrypt the bitstream. It is believed that every FPGA on the market as of mid-2012 was vulnerable to differential power analysis.

Other types of hardware tampering are more invasive. In some, called semi-invasive, only the package is removed. So the device can emit or respond to light but the silicon device is not mechanically disturbed. Even more invasive are attacks where the minuscule traces on an IC are cut or probed, or where the device is de-processed layer-by-layer to determine how it works. Generally, as an attack becomes more invasive, it is more expensive to mount, requiring expensive equipment like electron microscopes and focused ion beam machines. These attacks are also harder to defend against.

Technology differences

There are two main technologies used for commercial FPGAs today: SRAM and Flash. In addition, anti-fuse FPGAs are still used in space applications, which require enhanced tolerance to background radiation.

SRAM FPGAs, including those from Xilinx and Altera, hold device configuration bits in volatile SRAM cells. These bits determine the logic functions that the FPGA performs. However, since the configuration memory is volatile, it forgets the configuration whenever power is removed, and must be reloaded from external sources after each power-up. In low-end devices, the configuration bitstream moves from the external non-volatile memory where it is held, to the SRAM FPGA, in unencrypted form. This provides virtually no security, as anyone with a storage scope or logic analyser can easily record the data. Also, it is often trivially easy to copy the external memory chip to create clones of the system.

To address some of these weaknesses, higher-end SRAM FPGAs utilise encryption of the configuration bitstream. In these, if the option is used, the bitstream is stored in encrypted form in the external non-volatile memory, and decrypted on the fly by

the FPGA upon loading each time. For this to work, at least the decryption key needs to be stored in non-volatile memory in the predominantly volatile FPGA. This is sometimes done using a few bits of one-time programmable anti-fuse memory, or by using a battery to keep part of the chip alive at all times. Of course, if the battery goes dead or is disconnected even for a few milliseconds, the key is forgotten and the system won't boot up.

In flash-based FPGAs, configuration bits are held in non-volatile flash memory. Thus the configuration needs to be loaded only once—typically during the board assembly process. However, since flash memory is also reprogrammable, flash FPGAs can be reconfigured multiple times over their lifetime, if desired.

Flash-based cSoC devices also have on-chip embedded non-volatile memory (eNVM) for holding the processor firmware securely on-chip. For secure field upgrades of the FPGA fabric configuration and/or the eNVM array, most flash FPGAs also offer built-in bitstream decryption functionality. Unlike SRAM FPGAs that use decryption on every power-up cycle, flash FPGAs need to exercise this feature only when doing an upgrade. For higher security, these and related features can be permanently disabled.

Trends in advanced security architecture and countermeasures

With attacks continuously improving, FPGA countermeasures also must advance to keep ahead. The threat environment is daunting, but all hope is not lost. No security is absolute, but with proper architecture and design, next-generation FPGAs will be significantly stronger than those currently on the market.

Next-generation FPGAs and cSoCs would incorporate DPA countermeasures for all built-in 'bitstream' cryptographic operations. Over time, it is expected that deployment of DPA countermeasures will become the norm in the FPGA industry, as is the case with micro-